

# SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

## ONLINE VOTING SYSTEM

### 1. Introduction

#### 1.1 Purpose

The Online Voting System is a web-based project developed using PHP and MySQL. The main purpose of this system is to make the voting process simple and convenient. Registered users can log in and cast their votes online, while the administrator can manage voters, candidates, elections, and view the results.

#### 1.2 Objectives

- To provide an easy way for users to vote online.
- To allow only registered users to participate in voting.
- To prevent a voter from voting more than once in an election.
- To store voter, candidate, and voting details safely in a database.
- To help the administrator manage elections and check the results easily.

#### 1.3 Scope

This project covers voter registration and login, viewing candidates, casting votes, managing elections and candidates, and counting votes automatically. It is mainly developed as an academic project/prototype to demonstrate how an online voting system works.

### 2. Overall Description

#### 2.1 Users

##### **Voter:**

- Register an account and log in.
- View available elections.
- View the list of candidates.
- Cast a vote.
- Log out after using the system.

##### **Administrator:**

- Log in to the admin panel.
- Manage voters and candidates.
- Create and manage elections.
- View voting results.

- Log out of the system.

## 2.2 Technologies Used

- Frontend: HTML, CSS, JavaScript
- Backend: PHP
- Database: MySQL
- Server: XAMPP / Apache
- Browser: Chrome, Edge, Firefox

## 3. Functional Requirements

The system should provide the following functions:

- User Registration: New users can create their voter account.
- Login: Registered voters and administrators can log in using their credentials.
- Election Management: The administrator can create and manage elections.
- Candidate Management: The administrator can add, update, or remove candidates.
- Voting: Voters can select a candidate and submit their vote.
- Vote Restriction: The system should make sure that a voter cannot vote more than once in the same election.
- Vote Counting: Votes should be counted automatically and the results should be displayed.
- Logout: Users can log out safely after completing their work.

## 4. Non-Functional Requirements

- Security: User passwords and other important information should be protected.
- Usability: The system should have a simple and user-friendly interface.
- Performance: The system should respond quickly to normal user requests.
- Reliability: Voting and user data should be stored correctly.
- Compatibility: The system should work properly on commonly used web browsers.
- Maintainability: The code and database should be organized so that the system can be improved later.

## 5. Database Requirements

The system will use a MySQL database to store and manage information. The main tables will be:

| Table      | Purpose                             |
|------------|-------------------------------------|
| Users      | Stores voter details                |
| Admin      | Stores administrator login details  |
| Elections  | Stores election information         |
| Candidates | Stores candidate details            |
| Votes      | Stores the votes submitted by users |

The Votes table will keep track of the voter, election, and selected candidate. It will also help the system check whether a voter has already voted in a particular election.

## 6. Basic System Workflow

The basic working process of the system will be:

**Register/Login → View Election → View Candidates → Select Candidate → Confirm Vote → Store Vote → Show Confirmation**

The administrator will have a separate section where they can manage elections, candidates, voters, and view the voting results.

## 7. System Constraints

- Only registered users will be allowed to vote.
- A voter can vote only once in a particular election.
- Only authorized administrators can access the admin section.
- PHP, MySQL, and the required server environment must be properly configured.
- The system requires an internet/network environment when deployed as a web application.

## 8. Future Enhancements

- OTP or email verification
- Two-factor authentication
- Graphs and charts for results
- PDF generation for election results
- Better mobile-responsive design
- Detailed activity and audit logs

## 9. Conclusion

The Online Voting System using PHP and MySQL is designed to make the voting process easier and more organized. It allows users to vote online while giving the administrator control over elections, candidates, and results. Through this project, concepts such as PHP web development, MySQL database management, user authentication, CRUD operations, and vote counting can be practically implemented.